Multi-Modal AI For Cancer Cell Behaviour And Drug Discovery

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Abstract— Traditional cancer drug discovery remains time consuming and inefficient due to reliance on isolated data modalities. Most contemporary AI-based approaches are unimodal, concentrating solely on either molecular properties or medical imaging. This constraint limits the ability to capture the complex relationships between tumour behaviour and drug response.

A multi-modal deep learning framework is introduced to integrate histopathology-derived tumour representations with molecular drug information. Histopathology images are processed to extract cellular and spatial features through nuclei detection and graph construction. Convolutional neural networks (CNNs) extract high-level visual features, while graph neural networks (GNNs) model cell–cell interactions to encode structural characteristics of tumour tissue. The resulting embeddings are fused and clustered to identify tumour phenotypes, categorized as indolent-like, moderate-like, and aggressive-like. Concurrently, drug molecules are represented as graphs and processed using a molecular GNN trained on IC50 data to generate bioactivity-aware embeddings. Drug sensitivity information is incorporated to compute subtype-specific compatibility scores, facilitating phenotype-guided drug ranking.

The proposed framework creates a single picture of how tumors look and how they respond to drugs. This shows that it can effectively separate tumor subtypes and consistently prioritize drugs. This approach provides a scalable foundation for multi-modal, phenotype-driven cancer drug discovery.

Keywords— *Cancer Drug Discovery, Histopathology, Graph Neural Networks, CNN, Tumour Phenotype, Drug Recommendation, ChEMBL, GDSC*

I. INTRODUCTION

The leading cause of death around the world is cancer; however, its specific types vary. The most prevalent form of the disease is breast cancer. Despite advances in cancer treatment, the choice of drugs that effectively treat a particular condition is complicated by heterogeneity of tumours and different responses to treatments. Traditional processes used for discovering new drugs involve lengthy procedures, high costs, and reliance on either computational approaches or experimental assays independently.

Recently, AI and DL technologies proved to be promising for biomedical application, specifically in the areas of image processing in histopathology and molecular modelling. CNNs are used in histopathology to automatically detect visual features in images and classify tumours. While being widely adopted in the area, these methods usually do not account for spatial arrangements of cells and interactions between them that provide information about aggressiveness of tumours.

GNNs are commonly used to model the structure and interactions of entities. With histopathology, cellular arrangements can be mapped to a graph representation where nuclei serve as nodes and the relationship between them serves as edges. Similarly, chemical compounds can be also considered as a graph in drug modelling, allowing GNNs to learn chemical information and estimate the value of IC50 of a particular molecule.

Still, most of the existing approaches in the field are based on the use of a single data modality, focusing either on tumour analysis or drug modelling. Therefore, there is a lack of a means to establish a connection between two domains mentioned above. Moreover, few works attempt to incorporate multiple types of data (i.e., histopathology image data and drug data) into the same learning process.

In order to bridge the gap described above and to enable precise identification of a therapy that would fit a particular tumour type, it is proposed to develop a multi-modal deep learning method for guided drug recommendation. The proposed framework would allow for analysing both morphological properties of tumour tissues using histopathology images and predicting a bioactivity score for a certain drug.

CNN would be used to estimate high-level visual features in images. GNNs would be applied to model cellular graph and estimate interaction between cells. At the same time, another GNN would be trained on the basis of molecular graphs in order to estimate IC50. After that, the obtained representations would be used to form tumour embeddings and cluster them according to similarities.

Moreover, drug sensitivity data would help identify a list of drugs sensitive to a particular tumour subtype. Thus, the resulting ranking of the drugs would be obtained in order to recommend the most suitable treatment option.

II. Related Work

Deep learning methods have been widely used in various cancer-related studies and applications. Particularly, the applications of the methods include histopathology images and molecular property prediction. For instance, the convolutional neural network (CNN) has shown high accuracy in predicting tumours, their grading and subtype classification based on learning visual hierarchies from the images of tissue samples. Yet, the methods usually function at the image level, considering tumours independently and being unable to model cellular interactions and tumour structure that are necessary to understand tumour heterogeneity.

To improve the models' ability to work with these features, researchers have proposed the use of the graph-based approach where cellular nuclei are modelled using nodes while edges stand for the relationships in space. Graph neural networks (GNN) help better model cellular interactions. Nevertheless, there are no attempts of applying this approach beyond classification.

Furthermore, there has been growing interest in the molecular graph learning. This approach enables modelling the graph representation of drug compounds and capturing their chemical structures and features in order to predict their biological activities and model interactions between drug compounds and targets. However, it does not include any tumour-specific characteristics, leaving out the differences in tumour behaviour.

Finally, there has been an attempt to use multimodal learning for incorporating heterogeneous data in the prediction process and analysing the relationships between multiple types of information, including genomics, molecular and clinical data. Unfortunately, there are very few works combining histopathology data in the process.

For this reason, the proposed method combines the models of tumour representation based on histopathology images with the molecular graphs and drug sensitivity data into one multimodal pipeline.

Table 1- COMPARISON OF EXISTING TECHNIQUE WITH PROPOSED SYSTEM

Title	Author (s)	Advantages	Limitations
Heterogeneous Graph-Based Modeling of Whole-Slide Histopathology Images	Chan *et al.* [1]	Explicitly models spatial and semantic relationships in histopathology images using heterogeneous graphs	Complex graph construction and high computational cost limit scalability
Graphormer DTI: Graph Transformer for Drug–Target Interaction Prediction	Gao *et al.* [2]	Effectively captures molecular structural features using graph transformers; strong DTI prediction performance	Computationally expensive; heavily dependent on graph representation quality
Deep Learning for Histopathological Image Analysis	Spanhol *et al.* [3]	Demonstrates effectiveness of CNNs for feature extraction from histopathology images	Does not model spatial or relational tissue context
Deep Learning for Digital Pathology: A Tutorial	Janowczyk and Madabhushi [4]	Shows benefits of transfer learning and pretrained CNNs on medical images	Focuses on image-level predictions without relational modeling
CNN–Graph	Zhou *et*	Combines	Requires

Hybrid Framework for Medical Image Analysis	*al.* [5]	CNN embeddings with graph-based context modeling	careful graph construction and parameter tuning
RedCDR: Dual Relation Distillation for Drug Response Prediction	Xu *et al.* [11]	Integrates molecular graphs and omics data for improved drug response prediction	High model complexity and reliance on curated omics datasets
Proposed System	**This Work**	**Joint modeling of tumour behaviour and drug properties using CNNs and GNNs; modular and extensible design**	**Full multimodal fusion and large-scale clinical validation are future work**

III. Dataset Description

Cancer drug development workflows that have been conventionally practiced involve the use of monomodal datasets, including molecular features and biological activity tests. As such, the capability of these workflows in capturing the multi-scale relationship between tumor behaviour and the efficacy of drugs is limited. Importantly, these approaches are unable to account for tissue morphology and cellular distribution patterns, which are key aspects for understanding the aggressiveness of tumors and their responses to drugs. To circumvent this problem, the workflow to be developed will utilize a collection of heterogeneous data sets for representing various aspects of tumors, drugs and drug sensitivity.

The construction of the dataset will be based on a multi-modal approach, whereby the different types of data contribute to the respective stages of the workflow. For instance, histopathology images will be used to predict tumor behaviors, molecular bioactivity information will be applied to generate representations of drugs, and drug sensitivity data will be integrated to create phenotype-specific therapeutics.

A. Histopathology Image Dataset

The histopathology images are acquired from the BreaKHis (Breast Cancer Histopathological Database). This is a well-known benchmark dataset utilized extensively in computational pathology. In the dataset, there are images taken under the microscope for both benign and malignant tumours of breast tissues. These images are taken under four different magnifications of 40×, 100×, 200×, and 400×.

The use of histopathology images in this study is not for classification like conventional methods. Rather, it is used in tumor behavior modeling. As such, instead of predicting tumor class labels, this method uses biological features in order to model tumour behavior. Such biological features include density of cells, sizes of nuclei, and structural variations.

To incorporate spatial interaction features in the tumor modeling process, the nuclei are extracted from the images and modeled as graph nodes while their spatial relationship is captured as edges. Thus, graphs are developed which model the interactions between cells. The processed dataset is thus used in various downstream tasks, including nuclei detection, graph modeling, feature extraction through CNNs, and structural modeling through GNNs.

B. Molecular Bioactivity Dataset

The molecular data is extracted from the ChEMBL database, which is a manually curated knowledge base with the presence of chemical structures and bioactivity data with experimental validation. In this work, we have extracted about 5,000 drug-like molecules having IC50 data reported.

IC50 refers to the concentration required to inhibit the biological target by 50%, and it is one of the common measures to assess the potency of drugs. For ensuring consistent and reliable analysis, we have removed any entry having missing or invalid data in the IC50 column. All the IC50 values are also made in terms of nM.

Finally, each drug molecule is represented in terms of graph, where the nodes represent atoms and edges represent the bonding between these atoms. Such molecular graphs will be utilized for training the graph neural network so as to learn their latent representations.

C. Drug Sensitivity Dataset (GDSC)

To make a correlation between phenotype and drug sensitivity, information regarding drug sensitivity is retrieved from the Genomics of Drug Sensitivity in Cancer (GDSC) database. The

GDSC database holds information about IC50 values for different drugs tested on various cancer cells.

For this purpose, cell lines specific to breast cancer are considered, and mean IC50 values are calculated. These cell lines are then classified into three groups according to their sensitivity:

- Aggressive-like (high sensitivity / low IC50)
- Moderate-like
- Indolent-like (low sensitivity / high IC50)

The classification scheme matches the tumour phenotypes observed using histopathology data, allowing an indirect association of drug sensitivity to tumour subtypes.

After processing the GDSC data set, the compatibility score is calculated, which is then used for ranking of drugs during the final stage of our process.

D. Multi-Modal Dataset Integration

The three datasets collectively support the complete pipeline:

- Histopathology dataset → tumour representation (CNN + GNN)
- ChEMBL dataset → drug representation (molecular GNN)
- GDSC dataset → drug response mapping

Through the combination of these datasets, it provides an integrated approach to modeling tumour phenotypes and recommending drugs based on tumor phenotypes. This is something that cannot be achieved through the use of just one particular dataset.

IV. System Architecture

This study presents a novel framework that aims at overcoming the shortcomings of the existing unimodal methods in the process of cancer drug discovery. The majority of unimodal models consider either molecular or biological information alone, thus, ignoring the multilevel connection between tumour and drug. For these reasons, the proposed framework combines tumour visualisation by histopathology images and modelling of molecular drugs.

The overall system architecture includes two independent parallel pipelines, namely tumour analysis pipeline and molecular learning pipeline. As part of this architecture, curating and preprocessing of histopathology images and molecular bioactivity data occur first, followed by the operation of the two pipelines.

In tumour analysis pipeline, histopathology images are processed to capture the representation of tumour behaviour via nucleus identification, feature extraction, and cellular graph generation. Specifically, the use of CNN allows capturing high-level features, whereas GNN can be employed to model tumour microarchitecture using the generated cellular graphs.

Similarly, in molecular pipeline, molecular drugs are represented by means of molecular graphs, in which atoms and bonds act as nodes and edges correspondingly. GNN is used here in conjunction with IC50 bioactivity data for obtaining molecular embeddings.

Next, fusion of embeddings obtained by applying CNN and GNN to image and molecular graphs yields a tumour representation. Tumours are classified into indolent-like, moderate-like, and aggressive-like based on clustering, linking tumour analysis with drug discovery.

Drug effectiveness assessment relies upon inclusion of drug sensitivity data together with an IC50-based scoring scheme that helps rank drugs according to their effects on various tumour subtypes.

Overall, the framework comprises of four main parts: (i) representation learning of tumour behaviour with image processing and cellular graphs, (ii) feature extraction by CNN and GNN, (iii) embedding learning of molecules using GNN, and (iv) drug ranking based on clustering and IC50 scoring system.

V. Methodology

The methodology to be employed is comprised of several steps which aim at transforming unstructured information represented by histopathology images as well as molecular bioactivity data into structured representations that allow applying multimodal learning. Specifically, each step targets particular domain-related features with preservation of biologically

relevant information associated with the behaviour of tumours as well as drugs' activity.

A. Image Preprocessing and Nuclei Detection

There are some variations in histopathology images caused by factors such as variations in staining process, light conditions, and image noise. To ensure that there are no differences in images, each image is resized and standardized, i.e., converted into gray scale format.

The application of Gaussian filter reduces noise in the image, whereas Otsu filtering enables automatic segmentation of foreground regions. Erosion and dilation are other morphological filters used in the identification of nuclear structures by eradicating any kind of artifacts. After these preprocessing stages, contour detection technique is employed to detect individual nuclei.

Quantitative information on tumor shape is derived by computing certain metrics based on identified nuclei including cell density, nucleus size, and variance in sizes of nuclei.

B. Cellular Graph Construction

To represent spatial organization, the nuclei detected are mapped onto nodes within a graph, with each node having a set of geometric attributes that include area, perimeter, and circularity. These features help to reflect the cellular morphology.

The edges between nodes are derived from spatial proximity, where nuclei are linked based on a fixed threshold distance from their centers. In this way, the graph captures the interactions within cells and the overall organization of the tissue.

C. CNN-Based Image Feature Extraction

While creating the graph simultaneously, extraction of deep visual features takes place via a convolutional neural network. In this process, the pre-trained ResNet-18 architecture is used to obtain high-level embeddings from the histopathology image. The output layer responsible for classifying objects is discarded in order to prevent task-oriented learning.

The resulting high-dimensional embeddings represent features like texture, morphology, and structure of the input image and are task-independent, which makes them good and ideal for incorporation into graph-based features.

D. Molecular Graph Construction

The drug molecules are transformed into graphs based on their SMILES strings with the help of RDKit. In these graphs, each atom will be depicted by a node while each chemical bond between them by an edge.

Each node in the graph will have atomic numbers as its features, helping the model learn about the atomic nature of each molecule. Validation with the help of RDKit helps eliminate any invalid molecules.

E. Molecular Graph Neural Network Training

Training of a GCN model is performed on molecular graphs, with the IC50 values being used as the supervised information. Message-passing between nodes is carried out in order to obtain a representation of the local environment as well as higher-order structural dependencies.

The node embeddings obtained from the above process are pooled globally by taking the average, giving graph embeddings. The GCN is then optimized using regression loss, which allows for bioactivity-aware representations to be learned.

F. Tumour Representation Fusion and Phenotype Modeling

Tumour representation results from fusing CNN-derived image embeddings and GNN-derived graph embeddings. The fused embedding represents both image information as well as the underlying structure of tumour cells.

In order to obtain tumour patterns, first, PCA will be used for dimensionality reduction, followed by K-Means Clustering. In terms of tumour sub-types, the tumours will be classified as being one of the following three sub-groups:

- Indolent-like
- Moderate-like
- Aggressive-like

K-means clustering provides the link between image analysis and treatment planning.

G. Drug Ranking and Compatibility Analysis

In order to determine the connection between tumour phenotypes and drug sensitivity, drug sensitivity data are taken into account. Subtype-specific compatibility scores are calculated using IC50 values.

$$\text{Score} = 1 - \frac{\text{IC50} - \text{min}}{\text{max} - \text{min}}$$

Ranking is done using the computed scores for the drugs within each tumour subtype, leading to the generation of phenotypic drug recommendations. This process marks the final step in connecting tumour representations to drugs.

VI. ALGORITHMS USED

The suggested algorithmic framework involves multiple algorithms for representing diverse biomedical data through structure-based and representation-oriented learning. Selection of these algorithms depends both on types of data as well as their functions in the entire process.

A. Convolutional Neural Networks (CNNs)

Convolutional neural networks are employed to obtain features from the histopathology images. The architecture used for this process is a pretrained ResNet-18 model, whose convolution and pooling layers are responsible for learning the representations of tissue structure.

The learned representations involve different levels of features including textures at a lower level and tissue structure at a higher level. In order to facilitate generalization, the output layer of this model is discarded such that the architecture can act as a feature encoder.

B. Graph Construction Algorithms

Graph construction is applied to represent information from cells and molecules in a graph form.

Cellular Graphs:

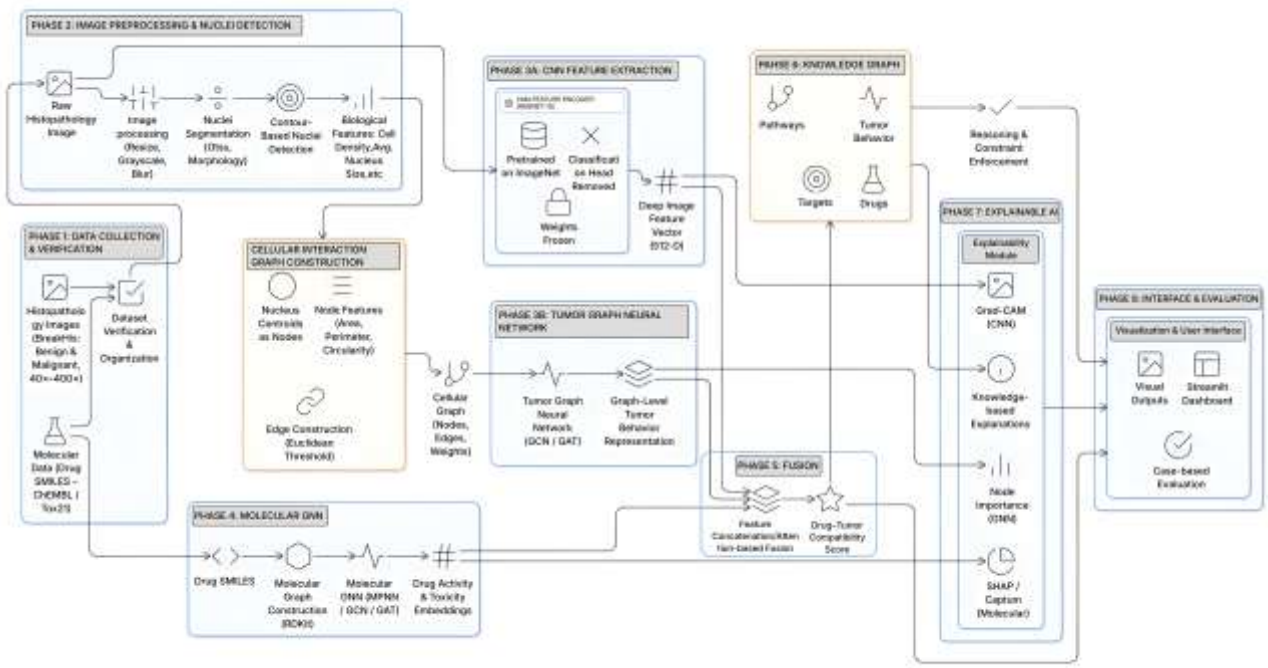
Cells obtained from histopathology images are represented as nodes whereas edges correspond to spatial closeness. Features of nodes are described by geometric descriptors including area, perimeter, and circularity. Such features are helpful for modeling tumor microstructure.

Molecular Graphs:

Graph representation is used for describing chemical compounds using SMILES strings where atoms are considered nodes and chemical bonds are edges. Such representation allows keeping the structure of molecules.

C. Graph Neural Networks (GNNs)

Graph Neural Networks are applied in order to learn the representations of cells and molecules. This can be later used for phenotype identification.



Cellular GNN:

It learns the structure of tumor tissue by pulling in information from nearby nuclei. Basically, this helps show how the tissue is organized and how different parts relate to each other spatially

Molecular GNN:

The graph convolutional network learns on the molecular graph data using IC50 values as guidance. The use of message-passing allows the model to comprehend the chemistry behind the structure dependencies between atoms. Global average pooling is used to learn graph-level embeddings of cellular and molecular graphs.

D. Clustering Algorithm (K-Means)

The K-Means algorithm can be applied to classify different phenotypes of tumors based on the combined embeddings from the two types of networks. Following the reduction of dimensions through PCA, the tumor data are then classified into three major clusters:

- Indolent-like
- Moderate-like
- Aggressive-like

This approach enables the identification of various types of variations within tumors without using any labeled subtypes.

E. Dimensionality Reduction (PCA)

The Principal Component Analysis (PCA) algorithm can be employed to reduce the dimensions of the tumor embeddings' concatenated vector, without losing the significant variability information contained within them. This facilitates clustering and enables visualization of tumor distribution in lower dimensions.

F. Drug Ranking Algorithm

Drug ranking uses IC50-based normalization to calculate compatibility scores:

$$\text{Score} = 1 - \frac{\text{IC50} - \text{min}}{\text{max} - \text{min}}$$

Drugs are ranked based on these scores for each tumor subtype, which makes it easier to choose better treatments that are more tailored to specific phenotypes.

VII. Implementation Details

This workflow was coded using modules in the Python programming language and Jupyter Notebook environment for facilitating modular development, experimentations, reproducibility, and transparency. Moreover, these modules can be reused for doing a similar kind of work in other multi-modal workflows.

The preprocessing and analysis of histopathology images were done through the use of OpenCV and NumPy libraries in dealing with high-resolution images. The preprocessing steps that include scaling, normalization, filtering, segmentation, and contouring are critical when constructing cellular graphs.

For the convolutional network architecture, the choice of PyTorch was made owing to its flexible nature, while PyTorch Geometric is utilized for implementing graph-based tasks including messaging and aggregation of cellular and molecular graphs. Lastly, the RDKit library was employed in parsing and standardizing molecules according to their SMILES representations.

Dimensionality reduction and k-means clustering were performed in Scikit-learn using principal component analysis. All the experiments were run in a GPU-driven workstation. Intermediate results, like images, feature vectors, graphs, embeddings, and clusterings were recorded during the process.

VIII. Experimental Results and Analysis

As observed in the results obtained from experiments, all components of the multi-modal framework function as expected. Image processing is able to produce precise and reliable image segmentation for various histopathology samples. Hence, cell detection and graph representation can be accurately generated and provide a reliable source for further steps of the process. As a result, variations in staining and imaging can be accounted for to ensure stability.
Regarding feature extraction, the CNN produces high-dimensional embeddings representing the most crucial morphological attributes, such as texture, spatial relations, and overall structure. At the same time, the cellular GNN learns structural representations emphasizing relationships within

a tissue. Together, they provide an informative representation of tumor dynamics.

Molecular GNN training converges steadily as a reaction to the selected loss function, i.e., IC50. In other words, it is capable of producing representations that correspond to the bioactivity of the analyzed chemicals. Their embeddings represent meaningful patterns, including bioactivity and structural properties of drugs.
Tumor representations' quality is estimated through PCA for dimensional reduction and K-Means for clustering into several categories based on the similarity of their properties. Thus, tumor samples cluster into three types: indolent-like, moderate-like, and aggressive-like tumors. Moreover, this classification is backed by the silhouette score obtained as a part of the analysis.

Incorporating information on drug sensitivity allows calculating compatibility of tumors and specific drugs. Based on the analysis of the ranking of drugs for each tumor subtype, one may observe significant differences in them: the aggressive subtypes are generally more sensitive to specific drugs compared to the others.

This pattern is also observable in the visualizations of embeddings and cellular graphs produced. Therefore, it appears reasonable to conclude that not only do the individual components of the framework perform efficiently; they also contribute to the development of an adequate tumor-drug relationship model.

IX. Discussion

Contrasted against previous drug discovery methodologies that consider only one kind of data, this approach integrates tumor morphology and drug molecular data. CNNs are used to extract image features, while graph-based techniques are applied to identify relationships in tumour structures, providing insight into not just the visual but also the relational nature of tumors and their treatment responses.

An essential aspect of the technique is the modeling of tumor microarchitecture using cellular graphs and the simultaneous learning of molecular characteristics using chemical structures. Thus, it considers not only spatial cell relationships but also biochemical properties, a feature that was absent in previous models since they treated each characteristic separately. Thus, this model can cluster tumours into phenotypes more accurately. Besides feature extraction, the method associate's tumour subtypes and drug sensitivity data and outputs clinically applicable results. In particular, it exhibits significant differences in the sensitivity of tumour phenotypes to drugs, indicating the necessity of personalized treatment methods.

However, there are some drawbacks of this method. Firstly, the size of the dataset is limited, which can lead to poor model performance due to the computation constraints. Moreover, tumour phenotypes are classified using unsupervised clustering without any clinical validation, decreasing confidence in its outcomes.

X. Conclusion and Future Work

This work looks at a multi-modal deep learning framework for cancer drug discovery, mainly focused on tumor phenotypes. The idea is to combine histopathology images with molecular graph data instead of treating them separately. CNNs are used to get features from the images, while GNNs are applied to both cellular and molecular data. So, in a way, the model is trying to capture multiple things at once—like tissue structure, how cells are positioned, and also the chemical side of drugs. Compared to more traditional approaches, this feels a bit more realistic since those usually focus on just one type of data and miss the bigger picture.

A key part of the framework is how it groups tumors into different phenotypes. It combines the features learned from CNNs and GNNs and then uses clustering to split them into categories like indolent-like, moderate-like, and aggressive-like. It's not a perfect system or anything, but it does give a rough way to distinguish between different tumor behaviors. After that, drug sensitivity data is used to rank drugs for each group, which makes the results more useful. It's not just analysis for the sake of it. It actually points toward possible treatment options.

The design is also pretty flexible. Everything isn't tightly connected, so parts like image processing or graph modeling can be changed without redoing the whole pipeline. That's helpful, especially if you want to improve certain components later or add new types of data. However, there are certain drawbacks. Since it is unsupervised clustering, there is no clinical

validation of these tumors, which gives a sense of ambiguity in results. Moreover, since it uses a small dataset, it is difficult to predict its performance in practical scenarios. It might require further improvement through additional information and clinical input.

The use of explainable AI can assist in improving the interpretation of the model since currently it is difficult to interpret the reasoning behind some of the predictions. Additionally, integration of biological knowledge graphs will assist in increasing the significance of the outcome. In summary, while it is a good effort, there is still much to improve on before implementation in the field.

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